

Pilbara Iron Strategic Decarbonization & Operational Excellence (PISDOE)

Strategic Project Management Portfolio | Tier-1 Mining Asset Integration

Executive Summary

The **PISDOE Project** is a strategic initiative designed to future-proof a Tier-1 Iron Ore asset in the Hamersley Basin, Western Australia. By integrating a **120MW Solar PV Array** and a **50MWh BESS (Battery Energy Storage System)**, the project aims to reduce Scope 1 & 2 emissions by **45%** while significantly lowering the **C1 Cash Cost** of production. This repository serves as a professional portfolio for a Data-Driven Project Manager, showcasing technical depth in financial modeling, risk mitigation, and operational transition within the Australian mining landscape.

Key Performance Indicators (KPIs)

Metric	Value	Technical Note
NPV (Net Present Value)	\$124M AUD	Calculated at 8% WACC over 12 years LOM.
IRR (Internal Rate of Return)	16.8%	Exceeds industry hurdle rates for green energy assets.
C1 Cost Reduction	-\$2.30 / tonne	Direct impact on operational margins and cutoff grades.
TRIFR Target	< 1.80	Aligned with global safety leadership benchmarks.

Technical Architecture & Standards

- **Energy Infrastructure:** 120MW Single-Axis Tracking (Bi-facial modules) optimized for Pilbara albedo.
- **Autonomous Integration:** Full software retrofit for **Autonomous Haulage Systems (AHS)** (Caterpillar/Komatsu).
- **Structural Compliance:** Designed for **Wind Region D** (Standard AS1170.2) to withstand severe cyclone activity.
- **Reporting Standard:** Resources and Reserves managed under **JORC Code 2012** guidelines for technical integrity.

Repository Structure

- **/index.html:** Interactive Digital Dashboard for real-time KPI visualization.
- **/Data:** Financial models including CAPEX/OPEX breakdown and Cash Flow simulations.
- **/Risk_Management:** 5x5 Matrix covering FIFO labor, seasonal weather, and Native Title (Heritage) risks.

- **/Docs:** Weekly Status Reports, Project Kick-off decks, and Closeout documentation.



Risk & Safety Leadership

Safety is non-negotiable in the Australian mining sector. This project prioritizes:

- **Zero Harm Culture:** Integrated safety audits during the pre-mobilization phase.
- **Heritage Protection:** Proactive engagement with Traditional Owners (Banjima/Yindjibarndi groups).
- **FIFO Logistics:** Robust management of Fly-In Fly-Out workforce requirements and remote area allowances.



Author

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Disclaimer: All data provided in this repository is simulated for professional portfolio purposes based on real-world mining benchmarks for the 2026 market outlook.